Notes: Boone & White (JFE, 2015)

The effect of institutional ownership on firm transparency and information production

Contents

1 Big picture

- **Question.** Does institutional ownership cause firms to become more transparent and generate more public information, or do institutions simply select firms that are already more transparent?
- **Empirical setting.** The paper exploits annual Russell 1000/2000 index reconstitution from 1996–2006. Near the cutoff, firms are similar in size and characteristics, but those at the top of the Russell 2000 receive much higher index weights than those at the bottom of the Russell 1000.
- **First-stage shock.** The Russell weighting rule creates a discontinuous increase in institutional ownership for Russell 2000 firms near the threshold. The increase is driven mainly by quasi-indexers.
- **Main outcome set.** The paper studies firm-produced public information, analyst information production, and the trading environment.
- **Main management-disclosure result.** Higher institutional ownership is associated with greater propensity to issue management forecasts, higher forecast frequency, more precise and timelier forecasts, and more total and voluntary 8-K filings.
- **Main analyst result.** Higher institutional ownership increases analyst following and lowers analyst forecast dispersion, suggesting that analysts respond to investor demand for information.
- **Main trading-environment result.** Higher institutional ownership lowers information asymmetry and increases liquidity: PIN and bid-ask spreads fall, while turnover and dollar volume rise.
- **Mechanism.** Quasi-indexers have large, diversified, long-term positions and value public information because it lowers monitoring and trading costs. Managers and analysts appear to respond to this information demand.
- **Economic takeaway.** Institutional owners, including quasi-indexers often viewed as passive, can produce positive information spillovers for all shareholders by inducing more transparency and improving market quality.

2 Incremental contribution of the paper

- **From association to quasi-causal evidence.** Earlier work documented correlations between institutional ownership and disclosure, analyst following, or liquidity. This paper uses Russell index reconstitution to isolate plausibly exogenous variation in institutional holdings.
- **Investor heterogeneity.** The paper does not treat institutions as homogeneous. It decomposes institutional ownership into quasi-indexers, dedicated institutions, and transient institutions. The economically important discontinuity is mainly quasi-indexer ownership.
- **Public-information channel.** The paper identifies firm transparency and analyst information production as concrete channels through which institutional investors affect the information environment.
- **Passivity is not irrelevance.** The findings qualify the view that passive or quasi-indexing institutions have weak incentives to monitor. Even when investors follow benchmark-like strategies, their demand for public information can shape corporate disclosure and analyst activity.
- **Link from governance to market quality.** The paper connects institutional ownership not only to disclosure choices but also to lower PIN, lower spreads, and higher trading activity, showing a capital-market payoff to information production.
- **Multiple robustness layers.** The paper combines a local RD design, pre-assignment balance tests, float-adjustment checks, an IV design using Russell weights and rankings, switcher analysis, and pre/post-Reg FD splits.

3 Data and measurement

3.1 Russell index reconstitution and treatment variation

Each year Russell Investments ranks exchange-traded U.S. common stocks by end-of-May market capitalization. The largest 1000 firms enter the Russell 1000, while firms ranked 1001–3000 enter the Russell 2000. Because both indexes are value-weighted, the smallest firms in the Russell 1000 receive small weights, while the largest firms in the Russell 2000 receive large weights relative to the index total.

Interpretation: This mechanical difference in benchmark weights creates the institutional ownership shock. Firms just above and just below the threshold are close in size, but they face very different demand from index-oriented institutions.

3.2 Institutional ownership and investor types

Institutional ownership is measured from SEC 13-F filings and is the percentage of shares held by institutions with at least $100 million in equity securities. The paper decomposes ownership using the Bushee classification:

- **Quasi-indexers:** low turnover, high diversification, long horizon; likely to value public, low-cost information because they hold broad portfolios.
- **Dedicated investors:** low turnover, low diversification, long horizon; concentrated holdings and potentially private monitoring.
- **Transient investors:** high turnover, high diversification, short horizon; may prefer frequent public information but can also impose disclosure-related concerns for managers.

Interpretation: The decomposition is central because the Russell shock is strongest for quasi-indexers. This allows the paper to interpret the results as evidence about indexing and benchmark-driven institutional ownership.

3.3 Firm transparency and management disclosure variables

Management disclosure is measured during the year after June index reconstitution, defined as July through the following May. The variables include:

- indicator for whether the firm provides any management forecast;
- forecast frequency;
- quarterly and annual forecast precision;
- quarterly and annual forecast horizon;
- total 8-K filing frequency;
- voluntary 8-K filing frequency.

Interpretation: These variables capture both whether managers disclose and whether disclosed information is more useful: more precise forecasts and earlier forecasts are more informative.

3.4 Analyst information production variables

Analyst data come from I/B/E/S. The paper focuses on one-year-ahead annual earnings forecasts. The main variables are:

- analyst following, measured as the number of unique analysts providing forecasts;
- analyst forecast dispersion, measured as the standard deviation of forecasts scaled by the stock price.

Interpretation: If institutional investors demand more public information, analysts should have greater incentives to cover treated firms. Lower dispersion suggests more precise shared information or less disagreement.

3.5 Trading-environment variables

The paper studies information asymmetry and liquidity through:

- PIN, the probability of informed trade;
- bid-ask spread;
- turnover;
- dollar volume.

Interpretation: A richer public-information environment should reduce adverse selection and improve liquidity. Quasi-indexer trading can also directly improve liquidity because their trading is often liquidity-motivated rather than information-motivated.

4 Models and empirical specifications

4.1 Local regression discontinuity design

The core design compares firms near the Russell 1000/2000 threshold. Let $R2000_i$ be an indicator that firm i is assigned to the Russell 2000, and let $Distance_i$ denote distance from the threshold. The estimated local RD relation can be written generically as

$$Y_{it} = \alpha + \tau R2000_{it} + f_{-}(Distance_{it})\mathbf{1}\{Distance_{it} < 0\} + f_{+}(Distance_{it})\mathbf{1}\{Distance_{it} \geq 0\} + \varepsilon_{it}, \quad (1)$$

where τ is the treatment effect of assignment to the top of the Russell 2000.

Interpretation: The key identifying comparison is local: firms just below the cutoff are in the Russell 1000 with small index weights, while firms just above the cutoff are in the Russell 2000 with large index weights.

4.2 First-stage institutional ownership discontinuity

The first-stage outcome is institutional ownership:

$$\mathrm{IO}_{it} = \alpha + \tau R2000_{it} + \text{local polynomial in distance} + \varepsilon_{it}. \quad (2)$$

Interpretation: A large positive τ for quasi-indexer ownership validates the mechanism: Russell assignment changes investor composition. Dedicated ownership is not the main source of variation.

4.3 Pre-assignment balance test

The paper applies the same RD logic to pre-assignment firm characteristics:

$$X_{it}^{pre} = \alpha + \tau R2000_{it} + \text{local polynomial in distance} + \varepsilon_{it}. \quad (3)$$

Interpretation: Balance tests ask whether the treatment and control firms differ before treatment in ways that could independently affect transparency or information production.

4.4 Outcome RD for disclosure, analysts, and trading environment

For each outcome Y, the estimated treatment effect is

$$Y_{it}^{post} = \alpha + \tau R2000_{it} + \text{local polynomial in distance} + \varepsilon_{it}. \tag{4}$$

Outcomes include management forecasts, 8-K filings, analyst following, forecast dispersion, PIN, spreads, turnover, and dollar volume.

Interpretation: The coefficient τ measures the discontinuity in the outcome induced by being assigned to the top of the Russell 2000.

4.5 Instrumental variables robustness

Following related Russell-index work, the paper estimates an IV design. The first stage instruments quasi-indexer ownership using Russell assignment and ranking:

$$QIO = \alpha + \tau R2000 + \delta Ranking + \gamma(R2000 \times Ranking) + \beta Year + \varepsilon. \tag{1}$$

The second stage estimates

$$Y = \alpha_0 + \alpha_1 \widehat{QIO} + \delta Weight + \gamma R2000 + \theta Y_{t-1} + \beta Year + \eta. \tag{2}$$

Interpretation: This robustness check addresses concerns about float adjustments and local functional-form assumptions. It asks whether the same results arise when quasi-indexer ownership is instrumented rather than using sharp RD treatment directly.

4.6 Switcher and Reg FD analyses

The paper also studies firms that switch indexes near the threshold. It compares firms moving up to the Russell 1000 with firms moving down to the Russell 2000. In addition, it splits the sample into pre- and post-Regulation Fair Disclosure periods.

Interpretation: Switchers provide within-firm evidence that changes in institutional ownership coincide with changes in disclosure, analyst coverage, and liquidity. The Reg FD split asks whether public-disclosure regulation changes the strength of the institutional ownership channel.

5 Identification and empirical design

5.1 What identifies the main effects

- Russell index construction creates a discontinuity in index weights at the Russell 1000/2000 cutoff.
- Near the cutoff, firms are similar in size and pre-assignment characteristics.
- The higher index weights of top Russell 2000 firms lead quasi-indexers to hold larger positions.
- The paper compares outcomes in the year after reconstitution.

Interpretation: The identifying assumption is local continuity: absent the ownership shock, firms just above and just below the cutoff would have similar information and trading outcomes.

5.2 Why the design is credible

- The cutoff is based on Russell's transparent ranking process.
- Firms do not know the exact threshold in advance and cannot easily manipulate their rank.
- Table 2 shows pre-assignment characteristics are broadly balanced.
- Figure 4 addresses the concern that Russell's float adjustment creates discontinuities in market capitalization.
- Robustness checks use alternative bandwidths, an IV design, switchers, pseudo-thresholds, and Reg FD subsamples.

Interpretation: The paper's empirical strength is the combination of a strong first stage and many tests against the main sources of RD failure.

5.3 Limitations and interpretation

- The treatment is not random assignment by researchers; it is quasi-experimental.
- The results are local to firms near the Russell 1000/2000 threshold.
- Russell membership changes both ownership and trading demand, so direct liquidity effects and disclosure-driven effects can interact.
- Some private communications between firms and institutions are not observed.
- The paper shows that institutional ownership affects information production, but it does not separately quantify every monitoring channel.

Interpretation: The safest interpretation is that quasi-indexer ownership improves the public information environment and trading conditions for mid-size firms near the Russell threshold.

6 Tables and figures

6.1 Figure 1: Sample timeline of Russell 1000 and Russell 2000 index reconstitution

Read:

- The timeline starts with the May 31 ranking date, when Russell ranks eligible U.S. common stocks by market capitalization.
- Preliminary additions and deletions are announced in mid-June.
- Russell updates the lists before the final reconstitution date.
- The final reconstitution occurs after market close near the end of June, and final membership lists are published in early July.

Economic message: The timeline clarifies why the outcome window begins after reconstitution. The shock is not a continuous event; it is a discrete annual portfolio-weighting change.

6.2 Figure 2: End-of-June index weights around the Russell 1000/2000 threshold

Read:

- The graph shows a sharp jump in index weights exactly at the threshold.
- Firms at the bottom of the Russell 1000 have very small portfolio weights in that large index.
- Firms at the top of the Russell 2000 receive much larger portfolio weights because they are among the largest firms in the smaller index.
- This discontinuity creates the institutional ownership first stage.

Economic message: This figure is the design's economic foundation. The index-weight discontinuity causes quasi-indexers to hold more of top Russell 2000 firms.

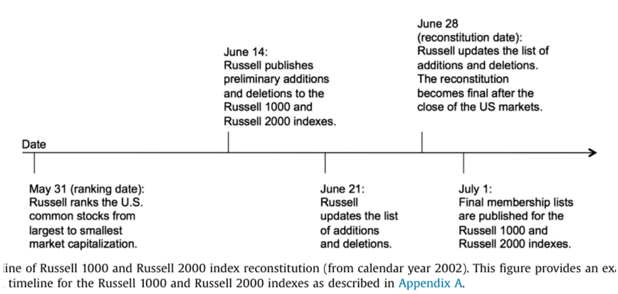

Figure 1: Sample timeline of Russell 1000 and Russell 2000 index reconstitution

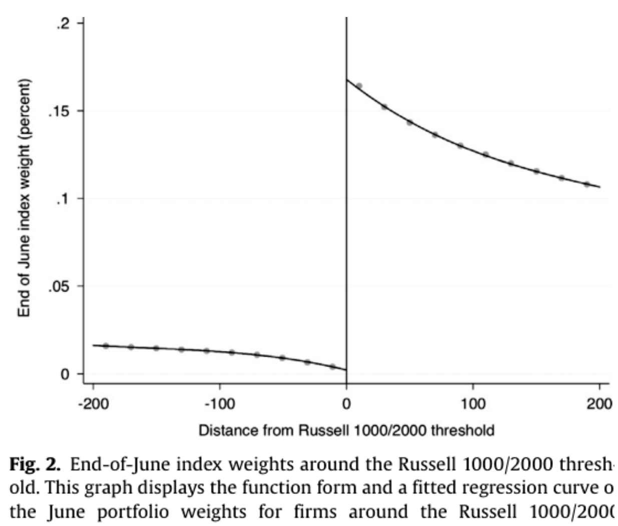

Figure 2: End-of-June index weights around the Russell 1000/2000 threshold

6.3 Figure 3: Institutional ownership around the Russell 1000/2000 threshold

Read:

- Panel A shows a discontinuity in total institutional ownership: firms just inside the Russell 2000 have much higher institutional ownership than firms just inside the Russell 1000.
- Panel B shows quasi-indexer ownership accounts for most of the jump.
- Panel C shows dedicated ownership is essentially smooth around the cutoff.
- Panel D shows transient ownership rises, but the jump is smaller than the quasi-indexer jump.

Economic message: The first stage is not generic institutional ownership. It is mainly quasi-indexer ownership, which supports the paper's interpretation that benchmark-driven investors increase public-information demand.

6.4 Table 1: Institutional ownership around the Russell 1000/2000 threshold

Read:

- Within the ± 50 bandwidth one quarter after reconstitution, total institutional ownership is 72.4% for top Russell 2000 firms versus 43.5% for bottom Russell 1000 firms.
- Quasi-indexer ownership is 42.6% versus 21.5%, a much larger gap than the dedicated-investor gap.
- The optimal-bandwidth RD coefficient for total institutional ownership in quarter $q+1$ is 0.392, implying a 39.2 percentage point discontinuity.
- The quasi-indexer RD coefficient is 0.261 in quarter $q+1$; dedicated ownership is not significantly affected.
- Effects persist in quarters $q+2$ and $q+3$, which matters because the outcomes are measured over the year after reconstitution.

Economic message: Table 1 quantifies the first stage and shows why the paper can study institutional ownership rather than merely Russell membership.

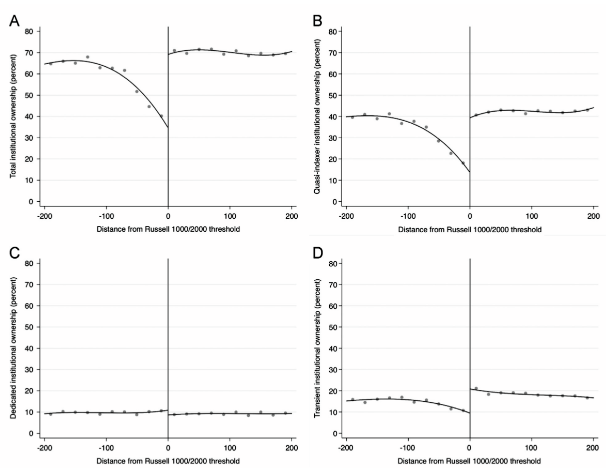

Fig. 3. Institutional ownership around the Russell 1000/2000 threshold one quarter (i.e., end of September) after the June index reconstitution. These graphs display the function form and a fitted regression curve of total (Panel A), quasi-indexer (Panel B), dedicated (Panel C), and transient institutional

Figure 3: Institutional ownership around the Russell 1000/2000 threshold

Table 1
Institutional ownership around the Russell 1000/2000 threshold.

This table compares institutional ownership during the sample period 1996–2006 for firms around the Russell 1000/2000 threshold for three quarters beginning in September following the June reconstitution of each index. Panel A compares the mean percentage of shares held by institutions for a set of three fixed bandwidths, where bandwidth is the number of firms on either side of Russell 1000/2000 threshold. Panel B presents the bias-corrected regression discontinuity (RD) treatment coefficient, τ, which represents the average treatment effect of assignment to the Russell 2000 index on institutional ownership. The RD coefficients are estimated by fitting a local third-order polynomial estimate using a triangular kernel to the left and right of the Russell 1000/2000 threshold using the bias-correction methodology in Calonico, Cattaneo, and Titiunik (2014a). We present the RD coefficients based on the rule of thumb bandwidth selection procedure prescribed in Calonico, Cattaneo, and Titiunik (2014b) and for two fixed bandwidths around the Russell 1000/2000 threshold. Institutional ownership is the percentage of shares outstanding held by institutions that hold at least $100 million in equity securities as reported on quarterly Securities and Exchange Commission Form 13-F filings. Institution types are from Brian Bushee's website (see http://acct.wharton.upenn.edu/faculty/bushee/IIclass.html) and are classified as quasi-indexer (low turnover, high diversification, long-term horizon); dedicated (low turnover and diversification, long-term horizon); and transient (high turnover and diversification, short-term horizon). [a], [b], and [c] indicate that the differences in mean values (Panel A) or the RD treatment coefficients (Panel B) are significantly different from zero at the 1%, 5%, and 10% levels, respectively. All variables are defined in Appendix B.

Panel A: Univariate analysis of institutional ownership (percent ownership of common shares)

	Bandwidth ± 50		Bandwidth ± 100		Bandwidth ± 200	
	Russell 1000	Russell 2000	Russell 1000	Russell 2000	Russell 1000	Russell 2000
Institutional ownership$_{q+1}$						
Total	43.5	72.4[a]	51.9	71.1[a]	58.6	70.3[a]
Quasi-indexer	21.5	42.6[a]	28.2	42.2[a]	33.8	42.4[a]
Dedicated	10.1	9.2	9.9	9.2	9.7	9.2
Transient	11.3	20.3[a]	13.1	19.4[a]	14.6	18.4[a]
Institutional ownership$_{q+2}$						
Total	45.9	72.0[a]	53.1	70.8[a]	59.5	70.1[a]
Quasi-indexer	22.1	41.5[a]	28.3	41.2[a]	33.6	41.7[a]
Dedicated	10.5	9.0[c]	9.9	9.0[c]	9.7	9.0[b]
Transient	11.8	20.0[a]	13.3	19.1[a]	14.7	18.0[a]
Institutional ownership$_{q+3}$						
Total	46.7	71.6[a]	53.5	70.4[a]	59.5	70.1[a]
Quasi-indexer	23.8	42.7[a]	29.6	42.3[a]	34.5	42.9[a]
Dedicated	10.8	9.2[c]	10.2	9.1[b]	9.9	9.1[a]
Transient	11.6	19.4[a]	13.1	18.6[a]	14.5	17.7[a]

Panel B: Regression discontinuity analysis of percentage institutional ownership

	Rule of thumb bandwidth			Fixed bandwidth	
	Treatment τ	z-Statistic	Bandwidth	Treatment τ Bandwidth ± 100	Treatment τ Bandwidth ± 200
Institutional ownership$_{q+1}$					
Total	0.392[a]	11.434	± 144	0.407[a]	0.400[a]
Quasi-indexer	0.261[a]	11.967	± 144	0.264[a]	0.282[a]
Dedicated	−0.008	−0.558	± 215	0.016	−0.007
Transient	0.126[a]	8.959	± 196	0.131[a]	0.128[a]
Institutional ownership$_{q+2}$					
Total	0.348[a]	9.749	± 150	0.363[a]	0.352[a]
Quasi-indexer	0.242[a]	11.044	± 146	0.249[a]	0.264[a]
Dedicated	−0.022	−1.408	± 214	0.007	−0.020
Transient	0.109[a]	6.821	± 178	0.107[a]	0.109[a]
Institutional ownership$_{q+3}$					
Total	0.291[a]	7.708	± 141	0.305[a]	0.316[a]
Quasi-indexer	0.217[a]	9.088	± 148	0.220[a]	0.241[a]
Dedicated	−0.026	−1.550	± 195	−0.006	−0.026[c]
Transient	0.098[a]	5.943	± 142	0.090[a]	0.096[a]

Table 1: Institutional ownership around the Russell 1000/2000 threshold

6.5 Figure 4: End-of-May market capitalization around the Russell 1000/2000 threshold

Read:

- Panel A shows that using the full sample can produce a visible discontinuity in end-of-May market capitalization.
- Panel B removes observations with large differences between predicted rank and actual June rank.
- After this adjustment, market capitalization appears smoother around the threshold.
- This addresses the concern that Russell's proprietary float adjustment may break local continuity.

Economic message: The figure is a validity check. If market capitalization were sharply different across the cutoff, outcome differences could reflect firm size rather than institutional ownership.

6.6 Table 2: Pre-assignment firm characteristics around the Russell 1000/2000 threshold

Read:

- The table compares size, cash-to-assets, ROA, book-to-market, earnings volatility, leverage, dividend payout, Tobin's q, and return volatility.
- The optimal-bandwidth RD coefficients in Panel B are generally small and statistically insignificant.
- The evidence supports balance in characteristics that prior literature links to disclosure and analyst coverage.

- Some fixed-bandwidth differences appear, but the main optimal-bandwidth RD tests support local comparability.

Economic message: Table 2 strengthens the RD interpretation: the discontinuity in information outcomes is unlikely to be driven by obvious pre-treatment differences.

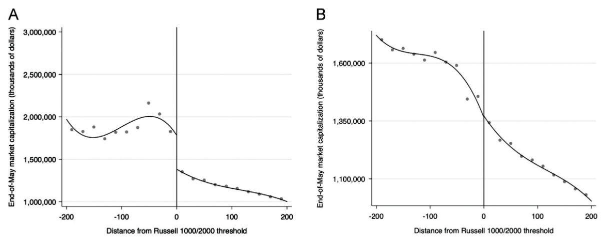

Fig. 4. End-of-May market capitalization around the Russell 1000/2000 threshold. These graphs display the function form and a fitted regression curve of

Figure 4: End-of-May market capitalization around the Russell 1000/2000 threshold

Table 2
Pre-assignment firm characteristics around the Russell 1000/2000 threshold.

This table compares pre-assignment firm characteristics during the sample period 1996–2006 for firms around the Russell 1000/2000 threshold. Panel A compares the sample mean values for each pre-assignment firm characteristic for a set of three fixed bandwidths, where bandwidth is the number of firms on either side of Russell 1000/2000 threshold. Panel B presents the bias-corrected regression discontinuity (RD) treatment coefficient, τ, which represents the average treatment effect of assignment to the Russell 2000 index on each baseline firm characteristic. The RD coefficients are estimated by fitting a local third-order polynomial estimate using a triangular kernel to the left and right of the Russell 1000/2000 threshold using the bias-correction methodology in Calonico, Cattaneo, and Titiunik (2014a). We present the RD coefficients based on the rule of thumb bandwidth selection procedure prescribed in Calonico, Cattaneo, and Titiunik (2014b) and for two fixed bandwidths around the Russell 1000/2000 threshold. [a], [b], and [c] indicate that the differences in mean values (Panel A) or the RD treatment coefficients (Panel B) are significantly different from zero at the 1%, 5%, and 10% levels, respectively. The definition and measurement period for all variables are described in Appendix B.

Panel A: Univariate analysis of baseline firm characteristics

	Bandwidth ± 50		Bandwidth ± 100		Bandwidth ± 200	
	Russell 1000	Russell 2000	Russell 1000	Russell 2000	Russell 1000	Russell 2000
Market capitalization (billions of dollars)	2.0	1.3[a]	1.9	1.2[a]	1.9	1.2[a]
Cash-to-assets (percent)	17.9	15.3[b]	16.5	14.3[b]	14.5	13.8
Return on assets (percent)	10.8	12.1	12.2	11.5	12.7	12.1
Book-to-market	0.4	0.4	0.4	0.5[b]	0.4	0.5[a]
Earnings volatility (percent)	88.9	10.3	90.2	19.2	49.0	18.9
Leverage (percent)	24.6	26.1	24.0	26.4[b]	24.2	25.8[b]
Dividend payout (percent)	8.7	7.7	7.4	7.8	7.2	7.6
Tobin's q	2.7	2.6	2.7	2.4[b]	2.5	2.2[a]
Return volatility (percent)	2.6	2.7	2.6	2.6	2.5	2.5[c]

Panel B: Regression discontinuity analysis of baseline firm characteristics

	Rule of thumb bandwidth			Fixed bandwidth	
	Treatment τ	z-Statistic	Bandwidth	Treatment τ Bandwidth ± 100	Treatment τ Bandwidth ± 200
Market capitalization	−0.179	−1.352	± 104	−0.165	−0.313[a]
Cash-to-assets	−0.021	−0.749	± 162	−0.027	−0.016
Return on assets	−0.006	−0.143	± 172	−0.057	0.009
Book-to-market	0.018	0.219	± 214	0.112	0.027
Earnings volatility	−3.919	−1.240	± 152	−4.975	−2.876
Leverage	0.006	0.177	± 173	0.033	0.002
Dividend payout	−0.033	−1.481	± 184	−0.002	−0.028
Tobin's q	0.664	0.227	± 196	−0.008	0.666
Return volatility	0.001	0.132	± 174	0.001	0.001

Table 2: Pre-assignment firm characteristics around the Russell 1000/2000 threshold

6.7 Table 3: Management disclosure around the Russell 1000/2000 threshold

Read:

- In the ±50 bandwidth, 52.73% of top Russell 2000 firms issue management forecasts versus 40.36% of bottom Russell 1000 firms.
- Forecast frequency is 1.81 versus 1.13 in the same narrow window.
- The RD estimate for being a forecasting firm is 0.139, and the estimate for forecast frequency is 0.504.
- Forecast precision and forecast horizon are also higher for treated firms.
- Total 8-K frequency and voluntary 8-K frequency rise by 1.447 and 0.720 in the optimal-bandwidth RD estimates.

Economic message: Higher institutional ownership changes managers' disclosure behavior on both extensive and intensive margins: more firms disclose, and disclosures are more frequent, precise, and timely.

6.8 Figure 5: Management disclosure around the Russell 1000/2000 threshold

Read:

- Panel A shows a jump in the probability of being a forecasting firm at the threshold.
- Panel B shows a jump in forecast frequency.
- Panels C and D show better quarterly forecast precision and earlier quarterly forecasts.
- Panels E and F show more total and voluntary 8-K filings.
- Several discontinuities reflect both an increase for treated Russell 2000 firms and a drop-off for bottom Russell 1000 firms.

Economic message: The graphical evidence complements Table 3 and makes the mechanism visually clear: institutional investor demand shifts the whole disclosure environment.

Table 3
Management disclosure around the Russell 1000/2000 threshold.

This table compares the properties of management disclosure during the sample period 1996–2006 for firms around the Russell 1000/2000 threshold. Management forecasts are measured during the year following index reconstitution, which we define as July through May following the June index reconstitution. Panel A compares the mean values of management disclosure properties for a set of three fixed bandwidths, where bandwidth is the number of firms on either side of Russell 1000/2000 threshold. Panel B presents the bias-corrected regression discontinuity (RD) treatment coefficient, τ, which represents the average treatment effect of assignment to the Russell 2000 index on the properties of management forecasts. The RD coefficients are estimated by fitting a local third-order polynomial estimate using a triangular kernel to the left and right of the Russell 1000/2000 threshold using the bias-correction methodology in Calonico, Cattaneo, and Titiunik (2014a). We present the RD coefficients are based on the rule of thumb bandwidth selection procedure prescribed in Calonico, Cattaneo, and Titiunik (2014b) and for two fixed bandwidths around the Russell 1000/2000 threshold. [a], [b], and [c] indicate that the differences in mean values (Panel A) or the RD treatment coefficients (Panel B) are significantly different from zero at the 1%, 5%, and 10% levels, respectively. All variables are defined in Appendix B.

Panel A: Univariate analysis of management disclosure

	Bandwidth ± 50		Bandwidth ± 100		Bandwidth ± 200	
	Russell 1000	Russell 2000	Russell 1000	Russell 2000	Russell 1000	Russell 2000
All management forecasts						
Forecasting firm (percent)	40.36	52.73[a]	44.09	50.27[a]	47.09	50.91[b]
Forecast frequency	1.13	1.81[a]	1.33	1.62[a]	1.49	1.58
Management earnings forecasts						
Quarterly forecast precision	2.04	2.08	2.01	2.06	2.02	2.05[c]
Annual forecast precision	2.04	2.12[b]	2.04	2.09[b]	2.04	2.08[b]
Quarterly forecast horizon (percent)	50.72	63.73[b]	54.67	57.29	59.23	54.80[c]
Annual forecast horizon (percent)	58.18	64.30[b]	59.71	62.51	59.79	61.56
Form 8-K filings						
Total 8-K frequency	5.44	6.52[a]	5.68	6.53[a]	6.16	6.47
Voluntary 8-K frequency	3.18	4.05[a]	3.33	3.94[a]	3.66	3.92[c]

Panel B: Regression discontinuity analysis of management disclosure

	Rule of thumb bandwidth			Fixed bandwidth	
	Treatment τ	*z*-Statistic	Bandwidth	Treatment τ Bandwidth ± 100	Treatment τ Bandwidth ± 200
All management forecasts					
Forecasting firm	0.139[b]	2.082	± 150	0.191[a]	0.185[a]
Forecast frequency	0.504[c]	1.706	± 147	0.668[b]	0.893[a]
Management earnings forecasts					
Quarterly forecast precision	0.349[a]	3.113	± 149	0.309[a]	0.291[a]
Annual forecast precision	0.262[a]	2.876	± 185	0.317[a]	0.268[a]
Quarterly forecast horizon	0.329[a]	2.740	± 179	0.383[a]	0.357[a]
Annual forecast horizon	0.172[a]	3.146	± 187	0.119[c]	0.166[a]
Form 8-K filings					
Total 8-K frequency	1.447[c]	1.858	± 149	2.312[a]	1.334[b]
Voluntary 8-K frequency	0.720[a]	2.813	± 168	1.411[a]	0.915[a]

Table 3: Management disclosure around the Russell 1000/2000 threshold

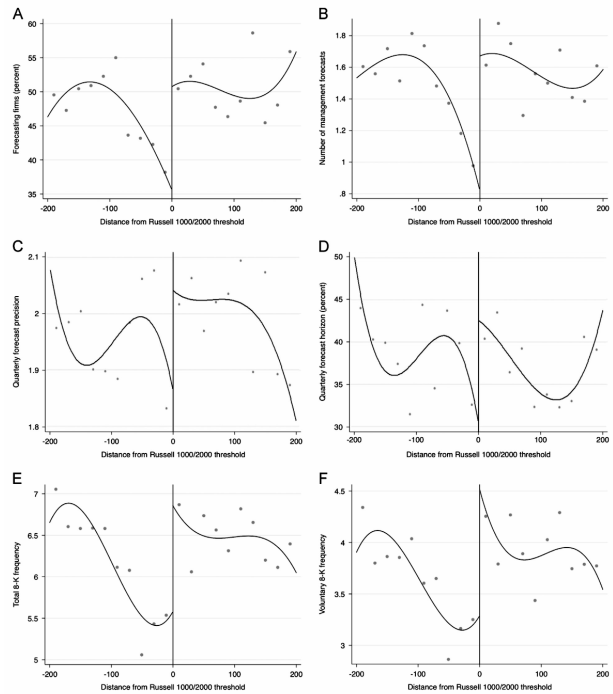

Figure 5: Management disclosure around the Russell 1000/2000 threshold

6.9 Table 4: Analyst forecasts around the Russell 1000/2000 threshold

Read:

- Analyst following rises for treated firms: RD estimates are 2.925 analysts at six months, 2.452 at 12 months, and 2.662 for the one-year average.
- Analyst forecast dispersion falls: the one-year average RD coefficient is -0.024.
- The strongest following result appears in the narrowest bandwidth, where size differences are minimized.
- Lower forecast dispersion suggests that the information set becomes more precise or more commonly shared among analysts.

Economic message: Institutional ownership affects not only firm-provided information but also outside information production by analysts.

6.10 Figure 6: Analyst forecasts around the Russell 1000/2000 threshold

Read:

- Panel A shows analyst following changes sharply around the threshold.
- The left side of the graph suggests a decline in coverage for bottom Russell 1000 firms as they move away from the threshold.
- Panel B shows analyst forecast dispersion is lower for treated firms.

- The figure supports the view that analysts allocate attention toward firms with greater institutional demand for information.

Economic message: Analysts appear to behave like information suppliers responding to clientele demand. More quasi-indexer ownership raises the value of analyst coverage.

Table 4
Analyst forecasts around the Russell 1000/2000 threshold.

This table compares the properties of analyst forecasts during the sample period 1996–2006 for firms around the Russell 1000/2000 threshold. Analyst forecasts are measured at six- and 12-month intervals as well as the one-year average following the index reconstitution in June. Panel A compares the mean values of analyst forecast properties for a set of three fixed bandwidths, where bandwidth is the number of firms on either side of Russell 1000/2000 threshold. Panel B presents the bias-corrected regression discontinuity (RD) treatment coefficient, τ, which represents the average treatment effect of assignment to the Russell 2000 index on the properties of analyst forecasts. The RD coefficients are estimated by fitting a local third-order polynomial estimate using a triangular kernel to the left and right of the Russell 1000/2000 threshold using the bias-correction methodology in Calonico, Cattaneo, and Titiunik (2014a). We present the RD coefficients based on the rule of thumb bandwidth selection procedure prescribed in Calonico, Cattaneo, and Titiunik (2014b) and for two fixed bandwidths around the Russell 1000/2000 threshold. [a], [b], and [c] indicate that the differences in mean values (Panel A) or the RD treatment coefficients (Panel B) are significantly different from zero at the 1%, 5%, and 10% levels, respectively. All variables are defined in Appendix B.

Panel A: Univariate analysis of analyst forecasts

	Bandwidth ± 50		Bandwidth ± 100		Bandwidth ± 200	
	Russell 1000	Russell 2000	Russell 1000	Russell 2000	Russell 1000	Russell 2000
Analyst following						
Six months after reconstitution	7.34	8.25[a]	7.81	8.05	8.67	7.76[a]
12 months after reconstitution	7.63	8.27[b]	7.86	8.06	8.64	7.72[a]
One-year average after reconstitution	7.26	8.25[a]	7.80	8.02	8.61	7.70[a]
Analyst forecast dispersion (percent)						
Six months after reconstitution	1.02	0.40	0.63	0.39	0.48	0.41
12 months after reconstitution	1.41	0.55[a]	1.67	0.74	1.74	0.62[c]
One-year average after reconstitution	1.38	0.66[a]	1.07	0.78	1.24	0.68[c]

Panel B: Regression discontinuity analysis of analyst forecasts

	Rule of thumb bandwidth			Fixed bandwidth	
	Treatment τ	z-Statistic	Bandwidth	Treatment τ Bandwidth ± 100	Treatment τ Bandwidth ± 200
Analyst following					
Six months after reconstitution	2.925[a]	4.653	± 180	3.350[a]	2.816[a]
12 months after reconstitution	2.452[a]	3.901	± 204	3.092[a]	2.460[a]
One-year average after reconstitution	2.662[a]	4.983	± 206	2.926[a]	2.669[a]
Analyst forecast dispersion					
Six months after reconstitution	−0.032	−1.555	± 143	−0.038[c]	−0.027[b]
12 months after reconstitution	−0.022[c]	−1.773	± 126	−0.021[b]	−0.016[c]
One-year average after reconstitution	−0.024[a]	−2.821	± 176	−0.027[a]	−0.025[a]

Table 4: Analyst forecasts around the Russell 1000/2000 threshold

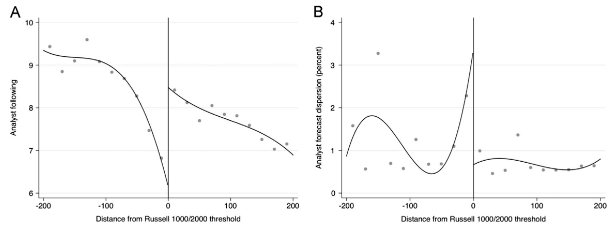

Fig. 6. Analyst forecasts around the Russell 1000/2000 threshold during the year after index reconstitution. These graphs display the function form and a fitted regression curve of analyst following (Panel A) and analyst forecast dispersion (Panel B) for firms around the Russell 1000/2000 threshold for the years 1996–2006. Each measure of analyst forecasts is defined in Appendix B. Distance represents the relative position of a firm centered on zero (the 1,000th firm) to the cutoff point between the indexes each year based on June weights. Negative values and zero represent the Russell 1000, and positive values represent the Russell 2000. The regression discontinuity plots represent local sample means using ten nonoverlapping evenly spaced bins on each side of the threshold following the methodology described in Calonico, Cattaneo, and Titiunik (2014a). The lines represent a third-order polynomial regression curve.

Figure 6: Analyst forecasts around the Russell 1000/2000 threshold

6.11 Table 5: Trading environment around the Russell 1000/2000 threshold

Read:

- In Panel B, the optimal-bandwidth RD coefficient for PIN is -0.049, indicating lower information asymmetry.
- Bid-ask spreads fall by -0.531 in the RD estimates.
- Turnover and dollar volume increase by 5.072 and 12.265, respectively.
- Panel C shows that within Russell 2000 firms, forecasting firms have lower PIN and spreads and higher turnover than non-forecasting firms.
- This suggests that management transparency partially accounts for the trading-environment improvement.

Economic message: The ownership shock improves market quality. It creates public-information spillovers that benefit more than the institutions that caused the change.

6.12 Figure 7: Trading environment around the Russell 1000/2000 threshold

Read:

- Panel A shows lower PIN on the Russell 2000 side of the threshold.
- Panel B shows lower bid-ask spreads on the Russell 2000 side.
- Panels C and D show higher turnover and dollar volume.
- Some of the visual discontinuity comes from a sharp worsening in the bottom Russell 1000, reinforcing the importance of the cutoff comparison.

Economic message: Greater institutional ownership improves the trading environment directly through liquidity demand and indirectly through better public information.

Table 5
Trading environment around the Russell 1000/2000 threshold.

This table compares proxies for information asymmetry and liquidity during the sample period 1996–2006 for firms around the Russell 1000/2000 threshold. The proxies are measured during the year following index reconstitution, which we define as July through May following the June index reconstitution. Panel A compares the mean values of each proxy for a set of three fixed bandwidths, where bandwidth is the number of firms on either side of Russell 1000/2000 threshold. Panel B presents the bias-corrected regression discontinuity (RD) treatment coefficient, τ, which represents the average treatment effect of assignment to the Russell 2000 index on each measure. The RD coefficients are estimated by fitting a local third-order polynomial estimate using a triangular kernel to the left and right of the Russell 1000/2000 threshold using the bias-correction methodology in Calonico, Cattaneo, and Titiunik (2014a). We present the RD coefficients based on the rule of thumb bandwidth selection procedure prescribed in Calonico, Cattaneo, and Titiunik (2014b) and for two fixed bandwidths around the Russell 1000/2000 threshold. Panel C compares the mean values of each proxy of the information environment for forecasting and non-forecasting firms in the Russell 2000. [a], [b], and [c] indicate that the differences in mean values (Panels A and C) or the RD treatment coefficients (Panel B) are significantly different from zero at the 1%, 5%, and 10% levels, respectively. All variables are defined in Appendix B.

Panel A: Univariate analysis of the trading environment

	Bandwidth ± 50		Bandwidth ± 100		Bandwidth ± 200	
	Russell 1000	Russell 2000	Russell 1000	Russell 2000	Russell 1000	Russell 2000
PIN	0.17	0.13[a]	0.16	0.14[a]	0.15	0.14[a]
Bid–ask spread	0.93	0.76[a]	0.85	0.79	0.82	0.82
Turnover	6.93	10.79[a]	7.65	10.36[a]	7.95	9.43[a]
Dollar volume (millions of dollars)	13.32	17.80[b]	14.93	16.01	15.47	13.22[a]

Panel B: Regression discontinuity analysis of the trading environment

	Rule of thumb bandwidth			Fixed bandwidth	
	Treatment τ	z-Statistic	Bandwidth	Treatment τ Bandwidth ± 100	Treatment τ Bandwidth ± 200
PIN	−0.049[a]	−5.623	± 128	−0.050[a]	−0.056[a]
Bid–ask spread	−0.531[a]	−4.094	± 162	−0.511[a]	−0.493[a]
Turnover	5.072[a]	3.225	± 135	5.638[a]	4.953[a]
Dollar volume (millions of dollars)	12.265[a]	3.572	± 197	17.328[a]	12.335[a]

Panel C: The trading environment by management forecasting decision

	Bandwidth + 50		Bandwidth + 100		Bandwidth + 200	
	No forecast	Forecast	No forecast	Forecast	No forecast	Forecast
PIN	0.14	0.12[a]	0.14	0.12[a]	0.15	0.13[a]
Bid–ask spread	0.96	0.60[a]	0.98	0.62[a]	0.97	0.68[a]
Turnover	9.86	11.55[c]	9.26	11.36[a]	8.35	10.39[a]
Dollar volume (millions of dollars)	17.99	17.64	15.45	16.51	12.42	13.93[c]

Table 5: Trading environment around the Russell 1000/2000 threshold

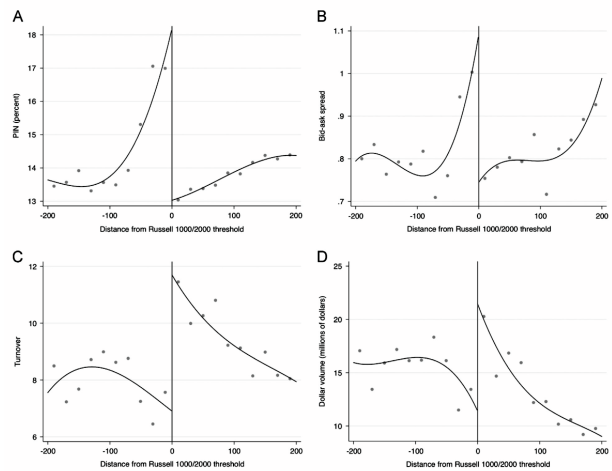

Figure 7: Trading environment around the Russell 1000/2000 threshold

6.13 Table 6: Instrumental variables analysis around the Russell 1000/2000 threshold

Read:

- The IV design instruments quasi-indexer ownership using Russell assignment, ranking, and index weights.
- Panel A shows instrumented quasi-indexer ownership raises management disclosure outcomes, including forecast precision, forecast horizon, and voluntary 8-K frequency.
- Panel B shows instrumented quasi-indexer ownership increases analyst following and lowers forecast dispersion.
- Panel C shows lower PIN, lower bid-ask spreads, and higher dollar volume.
- The IV results broadly match the RD results, reducing concern that the main findings reflect local functional-form choices.

Economic message: Table 6 supports the causal interpretation: when quasi-indexer ownership is isolated with instruments, the same information-production and liquidity patterns appear.

6.14 Table 7: Firms switching indexes around the Russell 1000/2000 threshold

Read:

- Firms moving down to the Russell 2000 experience larger increases in quasi-indexer ownership than firms moving up to the Russell 1000.
- Down-to-Russell-2000 firms show stronger increases in management forecasts, forecast frequency, forecast precision, forecast horizon, and 8-K frequency.
- Analyst following and trading-environment measures also improve in the predicted direction.

- This analysis uses within-firm changes around index switches, complementing the cross-sectional RD design.

Economic message: The switcher evidence shows that changes in institutional ownership are followed by changes in the information environment, not merely cross-sectional differences across firms.

Table 6
Instrumental variables analysis around the Russell 1000/2000 threshold.

This table presents the results of a two-stage least squares (2SLS) regression using an instrumental variable estimation based on Eqs. (1) and (2) from Section 5.1 for the fixed bandwidth of ± 100 around the Russell 1000/2000 threshold. The first stage estimates quasi-indexer institutional ownership (IO) as a function of the Russell 1000/2000 threshold. The second stage presents management disclosure (Panel A), analyst forecast (Panel B), and trading environment (Panel C) properties as a function of instrumented quasi-indexer institutional ownership. All regressions include year fixed effects and standard errors clustered at the firm level. In Panel A, forecast precision and horizon are presented for quarterly management earnings forecasts. ^{a}, ^{b}, and ^{c} indicate that the 2SLS regression coefficients (Panel C) are significantly different from zero at the 1%, 5%, and 10% levels, respectively. p-values are in parentheses. All variables are defined in Appendix B.

Panel A: Instrumental variables (2SLS) analysis of management disclosure

	Forecasting firm	*Ln(Forecast frequency)*	*Forecast precision*	*Forecast horizon*	*Ln(Total 8-K frequency)*	*Ln(Voluntary 8-K frequency)*
Instrumented quasi-indexer IO	0.182^{c}	0.244^{c}	0.395^{b}	0.497^{c}	0.572	0.568^{a}
	(0.082)	(0.091)	(0.041)	(0.083)	(0.113)	(0.007)
Russell weight	0.114	0.109	−0.054	−0.018	0.258^{b}	0.316^{a}
	(0.117)	(0.171)	(0.180)	(0.906)	(0.024)	(0.005)
Russell 2000	−0.155	−0.161	0.032	−0.015	−0.249	−0.370^{b}
	(0.118)	(0.140)	(0.465)	(0.943)	(0.120)	(0.019)
Lagged dependent variable	0.557^{a}	0.732^{a}	0.279^{a}	0.629^{a}	0.275^{a}	0.345^{a}
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Number of observations	2,128	2,128	473	473	2,128	2,128
r^2	0.361	0.557	0.174	0.529	0.549	0.387

Panel B: Instrumental variables (2SLS) analysis of analyst forecasts

	Ln(Analyst following)	*Ln(Analyst forecast dispersion)*
Instrumented quasi-indexer IO	1.381^{a}	−0.168^{c}
	(0.000)	(0.080)
Russell weight	0.725	0.184
	(0.548)	(0.957)
Russell 2000	−0.136	0.020
	(0.398)	(0.965)
Lagged dependent variable	0.719^{a}	0.218^{b}
	(0.000)	(0.016)
Number of observations	2,074	1,446
r^2	0.585	0.126

Panel C: Instrumental variables (2SLS) analysis of the trading environment

	PIN	*Bid–ask spread*	*Ln(Turnover)*	*Ln(Dollar volume)*
Instrumented quasi-indexer IO	−0.100^{a}	−0.390^{c}	0.411	0.792^{b}
	(0.000)	(0.091)	(0.156)	(0.030)
Russell weight	−0.018^{b}	−0.101	0.076	0.430^{a}
	(0.016)	(0.217)	(0.213)	(0.000)
Russell 2000	0.025^{b}	0.113	−0.047	−0.465^{a}
	(0.019)	(0.331)	(0.596)	(0.005)
Lagged dependent variable	0.497^{a}	0.585^{a}	0.880^{a}	0.757^{a}
	(0.000)	(0.003)	(0.000)	(0.000)
Number of observations	2,088	1,874	1,874	1,874
r^2	0.649	0.382	0.865	0.710

Table 6: Instrumental variables analysis around the Russell 1000/2000 threshold

Table 7
Firms switching indexes around the Russell 1000/2000 threshold.

This table compares the change in mean values of institutional ownership, management disclosure, analyst forecasts, and properties of the trading environment for firms that switch between the Russell 1000 and 2000 indexes. We focus our analyses on firms that switch indexes within the fixed bandwidth of ± 200 around the Russell 1000/2000 index threshold. This subsample contains 477 firms that move up to the Russell 1000 index and 312 firms that move down to the Russell 2000 index. Change in institutional ownership is measured as the difference in September following index reconstitution less March prior to index reconstitution. Changes in management disclosure and the trading environment are measured as the difference in the year following index reconstitution less the year prior to index reconstitution, which we define as July through May in each period. Changes in analyst forecast properties are measured as the difference between six months after and one month prior to index reconstitution. ^{a}, ^{b}, and ^{c} indicate that the differences in mean values is significantly different from zero at the 1%, 5%, and 10% levels, respectively. All variables are defined in Appendix B.

	Bandwidth ± 50		Bandwidth ± 100		Bandwidth ± 200	
	Up to Russell 1000	Down to Russell 2000	Up to Russell 1000	Down to Russell 2000	Up to Russell 1000	Down to Russell 2000
Change in institutional ownership						
Total (percent)	1.76	2.96^{c}	2.45	4.06^{c}	1.32	3.98^{a}
Quasi-indexer (percent)	0.22	4.32^{b}	1.47	4.05^{a}	0.72	3.85^{a}
Dedicated (percent)	1.10	0.02^{c}	0.27	0.25	0.38	0.51
Transient (percent)	1.70	−0.57	0.11	0.41	0.45	0.13
Change in management disclosure						
Forecasting firm (percent)	−11.11	4.92^{c}	−1.52	5.34^{c}	−4.01	4.98^{a}
Forecast frequency	−0.55	0.36^{b}	−0.15	0.27^{a}	−0.17	0.30^{a}
Forecast precision	−0.04	0.28^{b}	−0.02	0.19^{a}	0.02	0.17^{a}
Forecast horizon (percent)	−0.02	0.21^{c}	0.02	0.19^{a}	0.04	0.14^{b}
Total 8-K frequency	1.04	1.65^{c}	1.52	1.79^{c}	1.55	1.64
Voluntary 8-K frequency	0.61	1.13^{c}	0.81	1.11^{b}	0.93	1.04^{c}
Change in analyst forecasts						
Analyst following	0.22	0.32	−0.25	0.50^{a}	−0.48	0.44^{a}
Analyst forecast dispersion (percent)	−0.12	−0.26	−0.11	−0.21	−0.17	−0.74
Change in trading environment						
PIN (percent)	0.33	−1.57^{c}	0.01	−1.44^{a}	−0.21	−0.95^{a}
Bid–ask spread	−0.29	−0.19	−0.11	−0.22^{c}	−0.10	−0.25^{a}
Trading volume	−1.88	3.69^{b}	0.92	3.28^{b}	1.64	3.05^{b}
Turnover	−1.11	0.66^{c}	0.70	1.14^{c}	0.74	1.18^{c}

Table 7: Firms switching indexes around the Russell 1000/2000 threshold

6.15 Table 8: RD analysis around the Russell 1000/2000 threshold before and after Regulation Fair Disclosure

Read:

- The ownership first stage is present in both the pre-Reg FD and post-Reg FD periods.
- Management disclosure effects are much stronger post-Reg FD: forecasting firm, forecast frequency, forecast horizon, total 8-Ks, and voluntary 8-Ks are all positive.
- Analyst following rises and analyst forecast dispersion falls in both periods.
- PIN falls and liquidity improves in both periods, though individual measures vary in significance.
- The stronger post-Reg FD disclosure effects are consistent with a public-disclosure channel becoming more important after selective disclosure restrictions.

Economic message: Table 8 shows the ownership-information channel is not confined to one regulatory period. Reg FD appears to strengthen the public disclosure mechanism.

7 Short summary

- The paper studies whether institutional ownership causes greater firm transparency and information production.

Table 8
Regression discontinuity (RD) analysis around the Russell 1000/2000 threshold before and after Regulation Fair Disclosure.

This table presents regression discontinuity results for measures of institutional ownership, management forecasts, analyst forecasts, and proxies for information asymmetry, liquidity, and return volatility around Regulation Fair Disclosure (Reg FD), which became effective October 23, 2000. We define years 1996–1999 as pre-Reg FD and 2001–2006 as post-Reg FD. Institutional ownership is measured during the first quarter beginning September after the June index reconstitution. All other measures are averaged over the year following index reconstitution, which we define as July through May after the June index reconstitution. Management forecast precision and horizon are presented for quarterly management earnings forecasts. We present the bias-corrected RD treatment coefficient, τ, which represents the average treatment effect of assignment to the Russell 2000 index on each measure before and after Reg FD. The RD coefficients are estimated by fitting a local third-order polynomial estimate using a triangular kernel to the left and right of the Russell 1000/2000 threshold using the bias-correction methodology in Calonico, Cattaneo, and Titiunik (2014a). The reported RD coefficients are based on the rule of thumb bandwidth selection procedure prescribed in Calonico, Cattaneo, and Titiunik (2014b), where bandwidth is the number of firms on either side of Russell 1000/2000 threshold. [a], [b], and [c] indicate that the RD treatment coefficient is significantly different from zero at the 1%, 5%, and 10% levels, respectively. All variables are defined in Appendix B.

	Pre-Reg FD		Post-Reg FD		Full sample	
	Treatment τ	z-Statistic	Treatment τ	z-Statistic	Treatment τ	z-Statistic
Institutional ownership						
Total	0.428[a]	8.254	0.368[a]	8.979	0.392[a]	11.434
Quasi-indexer	0.211[a]	6.294	0.312[a]	12.563	0.261[a]	11.967
Dedicated	0.063[a]	3.197	−0.037	−1.452	−0.008	−0.558
Transient	0.138[a]	5.397	0.099[a]	5.655	0.126[a]	8.959
Management disclosure						
Forecasting firm	0.068	0.790	0.308[a]	3.665	0.139[b]	2.082
Forecast frequency	0.177	1.045	0.897[c]	1.813	0.504[c]	1.706
Forecast precision	2.491[a]	4.177	0.187[b]	2.002	0.349[a]	3.113
Forecast horizon	0.079	0.427	0.506[a]	3.706	0.329[a]	2.740
Total 8-K frequency	0.968	0.892	1.583[b]	2.373	1.447[c]	1.858
Voluntary 8-K frequency	0.438	0.488	0.947[a]	3.225	0.720[a]	2.813
Analyst forecasts						
Analyst following	3.648[a]	4.359	1.927[a]	2.562	2.662[a]	4.983
Analyst forecast dispersion	−0.035[b]	−2.045	−0.018[c]	−1.798	−0.024[a]	−2.821
Trading environment						
PIN	−0.063[a]	−4.282	−0.040[a]	−4.369	−0.049[a]	−5.623
Bid–ask spread	−1.053[a]	−4.420	−0.088	−1.302	−0.531[a]	−4.094
Turnover	4.585	1.626	2.887[c]	1.679	5.072[a]	3.225
Dollar volume	5.949	0.685	13.777[a]	3.634	12.265[a]	3.572

is unambiguously beneficial, it appears to have positive spillover effects in the Russell reconstitution setting.

indexes form the Russell 3000 index. Approximately two weeks after this date, Russell publishes a provisional list of

- It uses Russell 1000/2000 reconstitution to generate a discontinuity in institutional ownership near the cutoff.
- Top Russell 2000 firms receive higher index weights and have much higher institutional ownership than bottom Russell 1000 firms.
- The ownership jump is driven mainly by quasi-indexers.
- Pre-assignment firm characteristics are broadly balanced across the threshold.
- Higher institutional ownership raises management forecast propensity, forecast frequency, forecast precision, forecast horizon, and 8-K disclosure frequency.
- Higher institutional ownership raises analyst following and lowers analyst forecast dispersion.
- Higher institutional ownership lowers PIN and spreads and raises turnover and dollar volume.
- IV, switching-index, and Reg FD analyses support the robustness of the results.
- The main mechanism is that quasi-indexers prefer more public information because it lowers monitoring and trading costs.
- The key contribution is showing that institutional investors, even quasi-indexers, can create positive information spillovers for the market.

8 Conclusion

8.1 Core conclusion

Boone and White show that institutional ownership can causally improve a firm's public information environment. Using the Russell 1000/2000 reconstitution setting, they find that firms with exogenously higher quasi-indexer ownership provide more voluntary disclosure, receive more analyst attention, and trade in a more liquid environment with lower information asymmetry.

8.2 Economic mechanism

The mechanism is a demand-for-information channel. Quasi-indexers hold large, diversified, long-horizon portfolios. They cannot easily exit every low-information firm and may not gather costly private information about every holding. Public information is therefore valuable to them because it supports monitoring, proxy voting, valuation, and lower trading costs. Managers respond by supplying more forecasts and 8-K disclosures. Analysts respond by increasing coverage and reducing dispersion in forecasts.

8.3 Incremental contribution revisited

The paper's incremental contribution is to identify a quasi-experimental path from institutional ownership to transparency. It improves on prior correlational evidence by using Russell index assignment, and it clarifies that the relevant institutional owner is not the average institution but the quasi-indexer. It also widens the outcome set: institutions affect managerial disclosure, analyst coverage, forecast dispersion, PIN, spreads, turnover, and dollar volume.

8.4 Implications for passive ownership

The results imply that passive or quasi-indexing institutions need not be passive in their economic effects. They may not conduct firm-specific activism in every holding, but their demand for public information can influence managerial disclosure policies and analyst coverage incentives. This is a market-wide externality: improved public information benefits all shareholders, not just the institutions that generate the pressure.

8.5 Implications for disclosure and market quality

The paper connects corporate disclosure with secondary-market trading quality. More forecasts and more precise/timely disclosure reduce uncertainty and disagreement. Analysts also process and amplify information. The combined effect lowers adverse selection and increases liquidity. Thus, the transparency channel links governance, information production, and asset-market liquidity.

8.6 Limitations and caveats

The estimates are local to firms near the Russell 1000/2000 threshold and to the 1996–2006 sample period. The Russell setting generates a strong ownership shock, but it also changes trading demand and benchmark attention, so the direct and indirect channels are hard to fully separate. The paper observes public disclosure and analyst output but cannot observe every private engagement between managers and institutions. These caveats do not undermine the main result, but they matter for generalizing to very large firms, small firms, or other institutional investor settings.

8.7 Takeaway

The paper's broad takeaway is that ownership structure can shape information production. A change in who owns the firm changes the demand for transparency, which changes managerial disclosure, analyst behavior, and the trading environment. The Russell setting reveals a powerful empirical lesson: institutional investors can improve market information even when their ownership arises from indexing and benchmarking rather than discretionary stock picking.